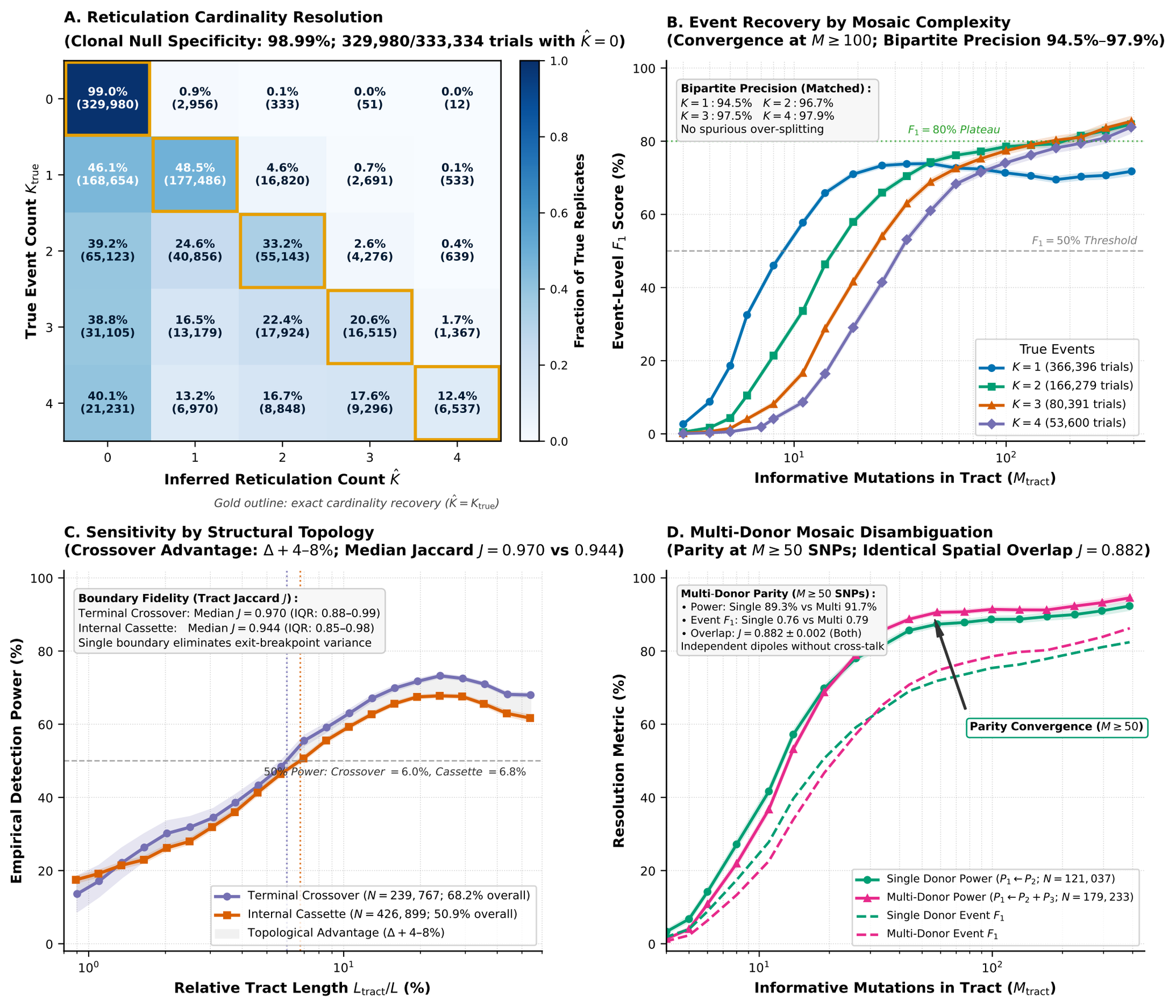

A. Reticulation Cardinality Resolution
(Clonal Null Specificity: 98.99%; 329,980/333,334 trials with $\hat{K} = 0$)
99.0% (329,980)
0.9% (2,956)
0.1% (333)
0.0% (51)
0.0% (12)
46.1% (168,654)
48.5% (177,486)
4.6% (16,820)
0.7% (2,691)
0.1% (533)
39.2% (65,123)
24.6% (40,856)
33.2% (55,143)
2.6% (4,276)
0.4% (639)
38.8% (31,105)
16.5% (13,179)
22.4% (17,924)
20.6% (16,515)
1.7% (1,367)
40.1% (21,231)
13.2% (6,970)
16.7% (8,848)
17.6% (9,296)
12.4% (6,537)
True Event Count K_{true}
Inferred Reticulation Count $\hat{K}$
Fraction of True Replicates
Gold outline: exact cardinality recovery ($\hat{K} = K_{true}$)
B. Event Recovery by Mosaic Complexity
(Convergence at $M \geq 100$; Bipartite Precision 94.5%–97.9%)
Bipartite Precision (Matched):
$K = 1$: 94.5% $K = 2$: 96.7%
$K = 3$: 97.5% $K = 4$: 97.9%
No spurious over-splitting
$F_1 = 80\%$ Plateau
$F_1 = 50\%$ Threshold
True Events
$K = 1$ (366,396 trials)
$K = 2$ (166,279 trials)
$K = 3$ (80,391 trials)
$K = 4$ (53,600 trials)
Event-Level F_1 Score (%)
Informative Mutations in Tract (M_{tract})
C. Sensitivity by Structural Topology
(Crossover Advantage: $\Delta + 4$–8%; Median Jaccard $J = 0.970$ vs 0.944)
Boundary Fidelity (Tract Jaccard J):
Terminal Crossover: Median $J = 0.970$ (IQR: 0.88–0.99)
Internal Cassette: Median $J = 0.944$ (IQR: 0.85–0.98)
Single boundary eliminates exit-breakpoint variance
50% Power: Crossover = 6.0%, Cassette = 6.8%
Terminal Crossover ($N = 239,767$; 68.2% overall)
Internal Cassette ($N = 426,899$; 50.9% overall)
Topological Advantage ($\Delta + 4$–8%)
Empirical Detection Power (%)
Relative Tract Length L_{tract}/L (%)
D. Multi-Donor Mosaic Disambiguation
(Parity at $M \geq 50$ SNPs; Identical Spatial Overlap $J = 0.882$)
Multi-Donor Parity ($M \geq 50$ SNPs):
• Power: Single 89.3% vs Multi 91.7%
• Event F_1: Single 0.76 vs Multi 0.79
• Overlap: $J = 0.882 \pm 0.002$ (Both)
Independent dipoles without cross-talk
Parity Convergence ($M \geq 50$)
Single Donor Power ($P_1 \leftarrow P_2$; $N = 121,037$)
Multi-Donor Power ($P_1 \leftarrow P_2 + P_3$; $N = 179,233$)
Single Donor Event F_1
Multi-Donor Event F_1
Resolution Metric (%)
Informative Mutations in Tract (M_{tract})